A BM EDITOR

bm Write: documents

Write letters and reports on the console, save them on the SD card and export them as PDF, HTML, Markdown or text.

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README

What this is. The manual of bm Write, the word processor of the bm suite.

How to use it. Read the first three chapters once, then follow the numbered steps of the tutorials in order. Anything in a bordered box with a shadow is meant to be copied as it is.

What is missing. Every gap is marked with a highlighted TO FILL tag; nothing is invented.

What bm Write is

bm Write is an app for writing documents. You type with a keyboard, or with the pad, and you see the text on A4 pages exactly as they will print.

Format	Extension	What it is
bm Write document	`.BMD`	the format you save and edit; plain text inside
PDF	`.PDF`	export, same lines on the same pages as the screen
HTML	`.HTM`	export
Markdown	`.MD`	export; it can also be opened
Text	`.TXT`	export; it can also be opened

What it does

- Eight paragraph styles: title, two headings, body, quote, bullet list, numbered list and table.
- Bold, italic and underline on single letters.
- Four alignments: left, centre, right and justify.
- Pages of A4 size with 2 cm margins and the page number at the bottom. The status line shows `Page 1 of 3` and the number of words.
- Word completion as in bm Code, in Italian or in English.
- Undo and redo, copy, cut and paste.

Why it matters

The PDF is made by the console itself. It has the same lines on the same pages as the screen, so what you see is what you print. The font is Courier, whose letters are all of the same width, as the screen's are.

Good to know

- Your documents live in the folder `/docs` on the SD card. They are shared by the apps that you allow to use them.
- The names are in the 8.3 form, without folders: up to 8 letters or digits, like `letter.bmd`.
- bm Write is a Market app: it is not part of the kernel.

First start

The question about documents

The first time an app uses your documents, the console asks you whether you allow it, as it does for the network. Your answer is kept in the file `bm/config.txt` on the SD card.

- If you say yes, bm Write can list, read, write and delete files in `/docs` .
- If you say no, saving, opening and exporting do not work: every call fails and the status line shows the reason.

The question reads "This document has changes. Save them?" and the buttons are Save, Discard and Cancel.

The first screen

You see an empty page on a dark desk. The cursor is a blinking bar. The toolbar at the top says `bm Write` , then `Untitled` . Start typing: the text appears on the page.

If you left a document with changes the last time, bm Write brings it back with the message `the document with its changes is back (Ctrl+S saves it)` . If you left a saved document, it opens it again.

> **NOTE**
> Press **F1** at any time to see the keys on the screen. Any key closes the list.

[TO FILL: shots/editor-write-start.png]

The screen and the keys

The screen

Part	Where	What it shows
Toolbar	top line	`bm Write`, the document name (`Untitled` if new, with `*` if it has changes), the style of the paragraph, the letters `B` `I` `U` and `L` `C` `R` `J`
Pages	the middle	the document on A4 pages, the cursor, the selection and the grey-blue word that Tab would write
Status line	bottom line	`Page N of M`, the number of words, `PAD` and the pad mode when pad typing is on, the language, and the chip F12 keys at the right (click it, or hold F12, to see every key of the page)

On the toolbar, the letters that are lit show the state at the cursor: bold, italic, underline, and the alignment of the paragraph. Messages such as `saved /docs/letter.bmd` take the place of the status line for a few seconds.

Writing and moving

Key	What it does
letters, digits, symbols	type at the cursor; a selection is replaced
Enter	new paragraph
Backspace, Del	delete backwards, forwards
Tab	write the grey-blue word; with no word, four spaces
Arrows, Home, End, PgUp, PgDn	move
F3	start or end a selection
Shift + arrows	select
Esc	with a selection: drop it; otherwise open the menu

Look and edit

Key	What it does
F2	next style of the paragraph
F4	next alignment of the paragraph
Ctrl+B, Ctrl+I, Ctrl+U	bold, italic, underline

Key	What it does
Ctrl+L, Ctrl+E, Ctrl+R, Ctrl+J	align left, centre, right, justify
Ctrl+Z, Ctrl+Y	undo, redo
Ctrl+C, Ctrl+X, Ctrl+V	copy, cut, paste
F1	the list of keys

Files

Key	What it does
Ctrl+S	save; asks for a name the first time
Ctrl+Shift+S	save as, with a new name
Ctrl+O	open a document from `/docs`
Ctrl+N	new document
Esc	the menu, with `Export...`

The menu

Press **Esc** (or **Start** on the pad). Use the up and down arrows and Enter. The entries are: New document, Open..., Save, Save as..., Export..., Style, Alignment, Bold, Italic, Underline, Select, Undo, Redo, Word completion, Language, Pad typing, Keys, Exit. `Word completion` switches the grey-blue words on and off. `Language` changes the language of the words between Italiano and English.

The mouse

A USB or Bluetooth mouse does what the keys do.

In the file name prompt, a click puts the caret between the letters.

Action	What it does
Click on the page	puts the cursor there
Drag	selects the text; at the top or bottom edge the page scrolls
Double click	selects the word
Shift + click	extends the selection
Wheel	scrolls the pages
Right button	opens the edit menu: Cut, Copy, Paste, Bold, Italic, Underline, Style, Alignment, Undo, Redo, Save, Menu
Click on `bm Write` in the bar	opens the menu
Click on the style, **B**, **I**, **U**, **L**, **C**, **R**, **J** in the bar	press the same key

In the menu and in the lists, a click chooses a row and a second click takes it. The wheel moves the choice. A click outside the box, or the right button, goes back.

The pad

Button	What it does
DPAD	move (it repeats if held)
confirm button	new paragraph
back button	delete
Start	the menu
Share	pad typing on or off

The key screen (F1) says "Share writes with the pad" and the menu says Share for pad typing; only the list shown while F12 is held says SELECT.

With the pad, A writes a new paragraph and B deletes; Select switches pad typing on and off (the menu calls it Share). **[TO FILL: check these buttons on the PowKiddy RGB30]**

While pad typing is on, the status line shows `PAD` and a small panel appears above it. **[TO FILL: how to compose letters with the pad]**

Styling

Paragraph styles

Press **F2** to go to the next style: Body, Title, Heading 1, Heading 2, Quote, Bullet list, Numbered list, Table, then Body again. The toolbar shows the name of the style.

Style	Look
Body	normal text, 72 letters per line
Title	large, bold, centred
Heading 1	large, bold
Heading 2	medium, bold
Quote	indented, italic
Bullet list	indented; one paragraph per item
Numbered list	indented; the numbers follow the paragraphs
Table	one row of a table; the bar `|` separates the cells

Tables

Choose the style Table (the last one in the **F2** list) and type a row with `|` between the cells, for example `Name | Price`. The bar becomes a thin line. The next row is another paragraph in the style Table. Press **Enter** at the end of a row to start a new row.

- Consecutive table rows share the widths of the columns. Each column is as wide as its longest cell. The cells line up.
- A row that is wider than 72 letters does not fit the page. bm Write shows it as body text, and the status line says `table too wide`. Make the cells shorter.
- Press **Enter** on an empty row, or **Backspace** at the start of a row, to change the row to body text.
- While the cursor is in a table row, the status line reminds you: `Table: | between the cells`.

Letters

Press **Ctrl+B**, **Ctrl+I** or **Ctrl+U** to switch bold, italic or underline.

- With a selection: the style is applied to the selected letters.
- With no selection: it applies to what you type next.

Alignment

Press **F4** to go through left, centre, right and justify, or press **Ctrl+L**, **Ctrl+E**, **Ctrl+R**, **Ctrl+J**. A justified paragraph widens the words to the right margin; its last line stays on the left.

Good to know

- Style and alignment belong to the whole paragraph. Bold, italic and underline belong to single letters.
- After a title or a heading, the next paragraph is Body. After a quote or a list item, the new paragraph keeps the same style; an empty list item becomes Body.

[TO FILL: shots/editor-write-typed.png]

Saving, opening, exporting

Where the files go

Everything goes in the folder `/docs` on the SD card. The folder is made by the first save. Take the card out and put it in your computer to find the files there.

Saving

1. Press **Ctrl+S**.
2. The first time, a line asks `Name of the document (8 letters, in /docs):` . Type up to 8 letters or digits (`-` and `_` are fine), for example `letter` . The name is written in lower case and `.bmd` is added.
3. Press **Enter**. The status line says `saved /docs/letter.bmd` , and the `*` next to the name goes away.

The next **Ctrl+S** saves with the same name. **Ctrl+Shift+S** asks for a new name.

Opening

1. Press **Ctrl+O**. If the document has changes, a box asks `This document has changes. Save them?` with Save, Discard, Cancel.
2. Pick a file from the list `Open a document (/docs)` with the up and down arrows. The list shows `.BMD` , `.MD` and `.TXT` files, with their size.
3. Press **Enter**.

A `.MD` or `.TXT` file opens as a new document with no name: **Ctrl+S** asks for a name and saves it as `.BMD` . Markdown marks are understood: `#` title, `##` and `###` headings, `>` quote, `-` bullets, `1.` numbers, `**bold**` , `*italic*` , `<u>underline</u>` . Rows with bars, such as `| a | b |` , come back as table rows; a row of the form `|---|---|` is skipped.

An older kernel does not know the Table style. It opens a saved table as body text, and the cells are separated by box-drawing bars.

Exporting

1. Press **Esc**, go down to `Export...` and press **Enter**.
2. In the box `Export to /docs as` , pick `PDF` , `HTML` , `Markdown` or `Text` , and press **Enter**.
3. If the document has no name yet, bm Write asks for one. The status line says `exported /docs/letter.pdf` with the size in bytes.

Format	File
PDF	`letter.pdf`
HTML	`letter.htm`
Markdown	`letter.md`
Text	`letter.txt`

An export is a copy to read or share. Save the `.BMD` too: it is the one you can edit with all the styles.

Tables go to each format in this way:

Format	A table
PDF	the same as on the screen
HTML	a `<table>`
Markdown	rows `| a | b |`, with a `|---|` row under the first row of the table
Text	the columns padded with spaces

BEWARE
An export with the same name replaces the older one in `/docs`.

Leaving

Choose `Exit` in the menu. If there are changes, a box asks you to save, discard or cancel. If you leave with Ctrl+Esc, the changes are asked about too; Ctrl+Esc a second time keeps them for next time.

Tutorials

Write and save your first document

> **TRY IT**
> Start at the console's menu, with the SD card in.

1. Start **bm Write** from the menu. You see an empty page and the toolbar `bm Write` `Untitled` .
2. Press **F2** once. The toolbar shows `Title` . Type `A letter` and press **Enter**. The title is large and centred.
3. The new paragraph is Body. Type `This text has a` then press **Ctrl+B**, type `bold` , press **Ctrl+B** again, and type `word.` The word `bold` is thicker.
4. Press **Enter**. Press **F2** five times to reach `Bullet list` . Type `one` , **Enter**, `two` , **Enter**, `three` . The three lines have bullets.
5. Press **Ctrl+S**. Type `LETTER` and press **Enter**.
6. Look at the status line: it says `saved /docs/letter.bmd` , and the toolbar shows `letter.bmd` with no `*` .

If the console cannot write, the status line is red and says `cannot save letter.bmd:` with the reason. See the last chapter.

[TO FILL: shots/editor-write-pages.png]

Export it as PDF and read it on your computer

> **TRY IT**
> Start with `letter.bmd` open and saved, as in the tutorial above.

1. Press **Esc**. The menu opens.
2. Go down to `Export...` and press **Enter**.
3. With `PDF` selected, press **Enter**.
4. Wait for `exported /docs/letter.pdf` on the status line.
5. Switch the console off and put the SD card in your computer.
6. Open the folder `docs` : `letter.pdf` is there. Open it with any PDF reader. The pages have the same lines as the screen, in Courier.

Open it again

> **TRY IT**
> Start with the document closed, or after starting bm Write again.

1. Press **Ctrl+O**. If bm Write asks to save changes, choose one.
2. The list `Open a document (/docs)` shows `letter.bmd` with its size, and the `.PDF` is not in the list.

3. Select `letter.bmd` with the arrows and press **Enter**.
4. The status line says `opened /docs/letter.bmd` . The text is as you left it.

You can also open `letter.md` , if you exported it: it comes back with the same paragraphs.

Tips, limits and troubleshooting

Tips

- Save early with **Ctrl+S**, and often.
- Use **F1** when you forget a key.
- Use **Tab** to take the grey-blue word instead of typing it all. Change the language in the menu if the words are not yours.
- With a selection, **Esc** drops the selection first; press it again for the menu.

Limits

- The name of a document is up to 8 letters or digits, `-` and `_` allowed, with no folders.
- A file written in `/docs` is at most 4 MiB.
- The body of a page holds 72 letters per line, in Courier.
- The text you did not save is kept for the next start only if it is under 24000 bytes.
- The title of the `.BMD` file keeps the first 60 letters of the first paragraph.
- **[TO FILL: the largest number of pages or paragraphs a document can have on the console]**
- The screen font is code page 437: Latin letters with accents, some Greek letters, box lines and signs of mathematics. The PDF uses the WinAnsi set (Latin letters with accents, curly quotes, the bullet and the euro sign). A sign that is not in WinAnsi, such as a box line, is written as `?` in the PDF.

Troubleshooting

Problem	What to do
Saving, opening or exporting fails and the status line is red	Read the reason. If you said no to the question on documents, the calls fail: the answer is kept in `bm/config.txt` on the SD card. Open that file on a computer and find the line that starts with `allow_` and ends with `(bm Write)`. Change `docs=no` to `docs=yes`, or delete the line: the console asks again at the next save
`no documents in /docs yet: Ctrl+S saves this one`	The folder is empty. Save a document first
`8 letters or digits, like LETTER`	The name has a space, or more than 8 letters (a final `.ext` is dropped: `letter.txt` is taken as `LETTER`). Type a shorter one
`not a bm Write document`	The `.BMD` file is not in the bm Write format
The file you saved is not in the open list	Only `.BMD`, `.MD` and `.TXT` are listed. A PDF or an HTML cannot be opened
You closed the app without saving	Start bm Write again: if the changes were kept you see `the document with its changes is back (Ctrl+S saves it)`
The words suggested are in the wrong language	In the menu choose `Language`; `Word completion` switches them off

NOTE

Check the PDF on the console before you rely on it for anything important: **[TO FILL: check the PDF on the Pi Zero W and open it on a PC, as the roadmap says]**

Open points

- **First start**: picture to take: shots/editor-write-start.png
- **The screen and the keys**: check these buttons on the PowKiddy RGB30
- **The screen and the keys**: how to compose letters with the pad
- **Styling**: picture to take: shots/editor-write-typed.png
- **Tutorials**: picture to take: shots/editor-write-pages.png
- **Tips, limits and troubleshooting**: the largest number of pages or paragraphs a document can have on the console
- **Tips, limits and troubleshooting**: check the PDF on the Pi Zero W and open it on a PC, as the roadmap says